Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$\begin{bmatrix}1\\3\end{bmatrix}$ m s^{-1}
			Total 1 mark

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an equation using conservation of momentum Condone sign errors with correct terms	AO1.1a	M1	CoM $4mu - mu = mv_A + 4mv_B$ $3u = v_A + 4v_B$ NLR $v_A - v_B = 2ue$ Subtracting equations gives $5v_B = 3u - 2ue$ $v_B = \frac{u(3-2e)}{5}$
	Obtains a correct momentum equation − can be unsimplified	AO1.1b	A1	
	Forms an equation using Newton's law of restitution	AO1.1b	B1	
	Completes a rigorous argument using both conservation of momentum and the coefficient of restitution to verify the correct speed of B	AO2.1	R1	
(b)	Substitutes the speed/velocity of B back into either of their equations	AO1.1a	M1	$v_B = \frac{u(3-2e)}{5}$
	Obtains the correct speed for A − must be fully simplified	AO1.1b	A1	
(c)	Uses the maximum and minimum values of e to consider the effect on the direction of motion	AO2.4	E1	Since $0 \le e \le 1$ then expressions for the speeds above are both positive
	Deduces that the spheres both travel in the same direction after the collision and justifies their conclusion	AO2.2a	R1	Hence the spheres both travel in the same direction
(d)	Recalls the formula for impulse and substitutes a pair of corresponding velocities into the impulse formula	AO1.1a	M1	$I = 4mv_B - 4mu_B$ $I = \frac{4mu(3-2e)}{5} - 4mu$ $I = \frac{8mu(1+e)}{5}$ $0 \le e \le 1$ $\frac{8mu}{5} \le I \le \frac{16mu}{5}$
	Substitutes 0 or 1 for e into 'their' expression	AO1.1a	M1	
	Completes a rigorous argument using algebraic expressions for the velocities, the formula for impulse and the range of values for e to verify the stated inequality.	AO2.1	R1	

Total 11 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses conservation of momentum to form an equation, with correct terms. (Condone sign errors.)	AO3.1b	M1	CoM: $0.8 \times 6 + 0.4 \times (-6)$ $= 0.8 \times (-1.2) + 0.4v_B$ $2.4 = -0.96 + 0.4v_B$
	Obtains correct velocity for *B* after the collision.	AO1.1b	A1	$v_B = 8.4$ Newton's law of restitution:
	Uses law of restitution to obtain an equation to find the velocity of *A*, with correct terms and 'their' velocity for *B*.	AO3.1b	M1	$-1.2 - 8.4 = -e(6 - (-6))$ $e = \frac{9.6}{12} = 0.8$
	Obtains a correct equation for 'their' values, follow through their other velocity, provided all M1 marks have been awarded.	AO1.1b	A1F	**AG**
	Completes a rigorous argument to reach the correct coefficient of restitution. **AG** Must use correct signs with the velocities.	AO2.1	R1	
(b)	States a valid refinement.	AO3.5c	E1	Include friction, which will reduce the speed of the discs as they move towards each other before the collision
				Total 6 marks

Q4.

Case 1: where 0.6 is taken as positive

$5 \times 4 - 4 \times 3 = 5 \times 0.6 + 4v$

$8 = 3 + 4v$

M1: Conservation of momentum, with left hand side as $5 \times 4 \pm 4 \times 3$.

A1: Correct equation (8 = 3 + 4v OE).

M1A1

v = 1.25 m s^{-1}

A1: Correct speed (1.25).

A1

Case 2: where 0.6 is taken as negative

$5 \times 4 - 4 \times 3 = 5 \times (-0.6) + 4v$

$8 = -3 + 4v$

M1: Seeing one of 8 = 3 – ± 4v or
– 8 = 3 ± 4v or 32 = – 3 ± 4v or
– 32 = 3 ± 4v OE

A1: Seeing ± 2.75 or $\frac{11}{4}$

M1A1

v = 2.75 m s^{-1}

A1: Correct speed. Accept $\frac{11}{4}$

A1

If mg used consistently instead of m deduct one mark, to give a maximum of 5 marks.

[6]

Q5.

$0.3 \times 2.8 = (0.3 + 0.2)\,v$

M1: Use of 2 or 3 term equation for conservation of momentum with 0.5 or equivalent on the RHS. Condone missing brackets if recovered.

A1: Correct equation.

M1A1

$v = \frac{0.3 \times 2.8}{0.5} = 1.68$ m s^{-1}

A1: Correct speed. CAO.

A1

Condone use of 300, 200 and 500 grams or use of correct ratios, eg 3, 2 and 5.

Note for consistent use of weight instead of mass penalise by one mark.

[3]

Q6.

(a) $m(4u) + 3m(2u) = mv_A + 3mv_B$

M1 for four correct momentum terms with any signs.
A1 for all correct

M1 A1

$\frac{v_B - v_A}{4u - 2u} = e$

$$\left(\begin{array}{l} v_A + 3v_B = 10u \\ v_B - v_A = 2ue \\ 4v_B = 2ue + 10u \end{array} \right)$$

M1 for correct terms for any signs, A1 for all correct.

M1 A1

$v_B = \frac{u}{2}(e + 5)$

OE, simplified

A1

$\left(v_A = \frac{u}{2}(e+5) - 2ue \right)$

$v_A = \frac{u}{2}(-3e + 5)$

OE, simplified

A1

6

(b) $e \le 1 \Rightarrow v_B \le \frac{u}{2}(1 + 5)$

Use of $e \le 1$ (OE) needed

FT their v_B

M1

$\Rightarrow v_B \le 3u$

A1

2

(c) $(I =)\ 3m.\frac{u}{2}(\frac{2}{3} + 5) - 3m.2u$

M1 for a difference of two momentums
FT their velocity from part (a)

M1

A1F for their 'Final B – Initial B'

A1F

$= \frac{5mu}{2}$ or $2.5mu$

A1

3

[11]

Q7.

$7(3\mathbf{i} + 8\mathbf{j}) + 3(6\mathbf{i} + 5\mathbf{j}) = 10\mathbf{v}$

M1: Three term equation for conservation of momentum with addition of terms and total mass of 10. Allow one error, for example switching masses or omitting negative sign in velocity.

A1: Correct equation for velocity.

M1A1

$\mathbf{v} = 3.9\mathbf{i} + 4.1\mathbf{j}$

A1: Correct velocity. Accept $\begin{bmatrix} 3.9 \\ 4.1 \end{bmatrix}$

A1

Finding speed as 5.66 without showing velocity scores M1 A0 A0.

Finding speed after having correct velocity should be considered as further work and not penalised.

Note: For consistent use of weight deduct one mark.

[3]

Q8.

$2 \times 4 + 3m = 3.8(2 + m)$

$8 + 3m = 7.6 + 3.8m$

$0.4 = 0.8m$

M1: Three term equation for conservation of momentum with correct RHS.
Allow 2 × 4 – 3m on the LHS
A1: Correct equation.

M1A1

$m = \frac{0.4}{0.8} = 0.5$ kg

A1: Correct answer.

A1

Note for consistent use of weight instead of mass penalise by one mark.

Allow use of any letter for the mass.

[3]

Q9.

$2 \times 4 + 3m = 3.8(2 + m)$

M1: Three term equation for conservation of momentum with correct RHS.

Allow 2 × 4 – 3m on the LHS

A1: Correct equation.

M1A1

$8 + 3m = 7.6 + 3.8m$

$0.4 = 0.8m$

$m = \frac{0.4}{0.8} = 0.5$ kg

A1: Correct answer.

A1

Note for consistent use of weight instead of mass penalise by one mark.

Allow use of any letter for the mass.

[3]

Q10.

$5 \times 6 = (m + 5) \times 2.4$

$30 = 2.4m + 12$

M1A1

M1: Equation for conservation of momentum with correct number of terms.

$$m = \frac{30 - 12}{2.4} = 7.5$$

A1: Correct equation.

A1: Correct mass CAO

Consistent use of weight instead of mass penalise final A1 mark.

A1

[3]

Q11.

(a) $6(5\mathbf{i} + 18\mathbf{j}) + m(2\mathbf{i} - 5\mathbf{j}) = 6(8\mathbf{i}) + m(V\mathbf{j})$

M1: Conservation of momentum, with addition of terms, as either 4 term vector equation (seen either in part (a) or part (b)) ***OR*** *three term equation for* ***i*** *component. Allow one error, for example switching masses.*

M1

$6 \times 5 + 2m = 6 \times 8$

$30 + 2m = 48$

A1: Correct equation for ***i*** *components.*

A1

$$m = \frac{48 - 30}{2} = 9$$

A1: Correct m.

A1

3

(b) $6 \times 18 - 5 \times 9 = 9V$

$108 - 45 = 9V$

M1: Conservation of momentum for ***j*** *component with correct signs.*

Allow one error, for example switching masses.

Note: omitting any mass scores M0.

A1F: Correct equation. Allow m instead of 9 at this stage.

M1A1F

$$V = \frac{108 - 45}{9} = 7$$

*A1F: Correct velocity. Condone 7**j***
FT candidate's mass from part (a).

Only award FT marks if mass positive.

Note $V = \dfrac{108}{m} = -5$

A1F

3

[6]

Q12.

$7 \times 10 + 3 \times 20 = 10v$

M1: Three term equation for conservation of momentum. Do not penalise inclusion of negative signs. Must see a combined mass of 10.

M1

A1: Correct equation. Accept 3v + 7v in place of 10v.

A1

$$v = \frac{130}{10} = 13\,\text{m s}^{-1}$$

A1: Correct speed.
Consistent use of mg instead m throughout deduct 1 mark.

A1

3

[3]

Q13.

(a) $$6\begin{bmatrix}2\\4\end{bmatrix} + m\begin{bmatrix}3\\-2\end{bmatrix} = 6\begin{bmatrix}1\\3\end{bmatrix} + m\begin{bmatrix}7\\b\end{bmatrix}$$

M1: Four term conservation of momentum equation. Allow sign errors.

M1

A1: Correct equation with correct signs. Vector equation may be implied by later correct working in this part of the question.

A1

$6 \times 2 + 3m = 6 \times 1 + 7m$

A1: Correct equation for correct component.

A1

$12 + 3m = 6 + 7m$
$6 = 4m$
$m = 1.5$

A1: Correct m.

A1

Example if only 12 + 3m = 6 – 7m without a vector equation award M1A0A0A0.

4

(b) $6 \times 4 + 1.5 \times (-2) = 6 \times 3 + 1.5b$

B1F: Correct equation using m or candidates m from (a).

B1F

$24 - 3 = 18 + 1.5b$
$3 = 1.5b$
$b = 2$

B1F: Correct b from candidate's m from (a).

B1F

Note: $b = \frac{6}{m} - 2$

2

Consistent use of mg instead of m throughout penalise 1 mark.

[6]

Q14.

(a) C.L.M.

$(1)3u = (1)v_A + (3)v_B$

M1 for three non-zero terms

M1A1

Restitution:

$\frac{1}{3} \times 3u = v_B - v_A$

Accept $v_A - v_B$

M1A1

$v_B = u$
$v_A = 0$

Solution

m1

A1 for both answers

A1

6

(b) C.L.M.

$3u = 3w_B + xw_c$

M1A1

Restitution:

$\frac{1}{3}u = w_c - w_B$

M1A1

$w_C = \frac{4u}{3 + x}$

Solution attempt, dep. on both M1s

AG

m1

$w_B = \frac{u(9-x)}{3(3+x)}$ OE

A1 for both

A1

6

(c) For further collision $\frac{u(9-x)}{3(3+x)} < 0$

M1

$9u - xu < 0$

$x > 9$

AG

A1

2

(d) $I = 5(\frac{4u}{3+5})$

M1

$I = \frac{5u}{2}$

A1

Alternative:

$I = 3u - 3 \times \frac{u(9-5)}{3(3+5)}$

(M1)

$I = \frac{5u}{2}$

Accept $-\frac{5u}{2}$

Follow through on their w_B

(A1F)

2

[16]

Q15.

(a) The spheres are smooth, no force acting in **j** direction

Any valid reason

E1

1

(b) $v_A = a\mathbf{i} + b\mathbf{j}$

$v_B = c\mathbf{i} + d\mathbf{j}$

C.L.M. along **i**: $1(2) + 2(-1) = 1(a) + 2(c)$

$a + 2c = 0$

M1A1

Restitution along **i**: $c - a = 0.5(2 - (-1))$

$c - a = 1.5$

$c = 0.5$

$a = -1$

M1A1

$v_A = -\mathbf{i} + 3\mathbf{j}$

A1F

$v_B = 0.5\mathbf{i} - 2\mathbf{j}$

A1F

6

[7]

Q16.

$4 \times 8 = (4 + 1)v$

M1: Three term momentum equation

M1

A1: Correct equation

A1

$v = \frac{32}{5} = 6.4 \text{ m s}^{-1}$

A1: Correct speed

A1

3

[3]

Q17.

$2.5 \times 12 + 1.5 \times 4 = 4v$

M1: Three term momentum equation,
correct values but condone incorrect signs.

M1

A1: Correct equation with correct signs.

A1

$v = \frac{36}{4} = 9$ m s^{-1}

A1: Correct speed
Note: Consistent use of mg instead of m throughout deduct 1 mark.

A1

3

[3]

Q18.

(a) $3\begin{bmatrix}6\\-2\end{bmatrix} + 7\begin{bmatrix}-1\\4\end{bmatrix} = 10\mathbf{v}$

M1: Forming three term equation for conservation of momentum, but condone incorrect signs. Must see combined mass of 10.

M1

A1: Correct equation with correct signs.

Accept $3\begin{bmatrix}6\\-2\end{bmatrix} + 7\begin{bmatrix}-1\\4\end{bmatrix} = 3\mathbf{v} + 7\mathbf{v}$ *oe*

A1

$\mathbf{v} = \frac{1}{10}\begin{bmatrix}11\\22\end{bmatrix} = \begin{bmatrix}1.1\\2.2\end{bmatrix}$

A1: Correct velocity
Consistent use of mg instead of m throughout deduct 1 mark

A1

3

(b) $v = \sqrt{1.1^2 + 2.2^2}$

M1: Finding speed. Must be + inside square root.

M1

$v = 2.46$ m s^{-1}

A1F: Correct speed for their velocity

Accept $1.1\sqrt{5}$ *or* $\frac{11\sqrt{5}}{10}$ *or 2.45 or AWRT 2.46*

A1F

2

[5]

Q19.

(a) $5mu + 7mu = mv_A + 7mv_B$

Allow consistent use of positive or negative sign for v_A.

M1A1

$12u = v_A + 7v_B$

$e = \dfrac{-v_A + v_B}{4u}$

M1

$-v_A + v_B = 4eu$

$8v_B = 12u + 4eu$

m1

$v_B = \dfrac{u}{2}(e + 3)$

AG

A1

5

(b) $v_A = \dfrac{u}{2}(e + 3) - 4eu$

M1

$v_A = \dfrac{u}{2}(3 - 7e)$

A1F

$\dfrac{u}{2}(3 - 7e) < 0$

M1

$3 - 7e < 0$

$e > \dfrac{3}{7}$

AG

A1

4

(c) $w_B = \frac{u}{4}(e + 3)$

M1

$\frac{u}{2}(7e - 3) < \frac{u}{4}(e + 3)$

M1

$2(7e - 3) < e + 3$
$13e < 9$

m1

$e < \frac{9}{13}$

AG

A1

4

[13]

Q20.

(a) $5\begin{bmatrix} 2U \\ U \end{bmatrix} + 15\begin{bmatrix} V \\ -1 \end{bmatrix} = 20\begin{bmatrix} V \\ 0 \end{bmatrix}$

Three term equation for conservation of momentum.

M1

$5U - 15 = 0$

Equation for U based on conservation of momentum.

dM1

$U = 3$

Correct value for U.
Deduct one mark for using weight instead of mass.

A1

3

(b) $30 + 15V = 20V$

Equation for V based on conservation of momentum.

M1

30 = 5V

$V = \frac{30}{5} = 6$

Correct value for V.
Deduct one mark for using weight instead of mass.

A1F

2

[5]

Q21.

(a) (i) $5\begin{bmatrix} 2U \\ U \end{bmatrix} + 15\begin{bmatrix} V \\ -1 \end{bmatrix} = 20\begin{bmatrix} V \\ 0 \end{bmatrix}$

Three term equation for conservation of momentum.

M1

$5U - 15 = 0$

Equation for U based on conservation of momentum.

dM1

$U = 3$

Correct value for U.
Deduct one mark for using weight instead of mass.

A1F

3

(ii) 30 + 15V = 20V

30 = 5V

Equation for V based on conservation of momentum.

M1

$V = \frac{30}{5} = 6$

Correct value for V.
Deduct one mark for using weight instead of mass.

A1F

2

(b) $v = \sqrt{3^2 + 6^2} = 3\sqrt{5} = 6.71\ \text{m s}^{-1}$

Calculation of speed.

M1

Correct speed.
Allow $\sqrt{45}$

A1F

2

[7]

Q22.

(a) 2*m* – 2 × 3 = *m* × (–0.5) + 3 × 0.5

Equation for conservation of momentum with four terms: 2m, 2 × 3, 0.5m and 3 × 0.5 regardless of signs.

M1

Correct equation with correct signs

A1

2.5*m* = 7.5
m = 3 kg

Correct mass
Arguments based on the symmetry of the situation that lead to m = 3 can be awarded full marks.
Note: Consistent use of mg instead of m: deduct one mark.
Note: Use of all positive signs leads to m = – 3, which might be changed to + 3 by candidates (M1A0A0).
Note: m = 3 can be obtained vis 1.5m = 4.5, which will usually score M1A0A0.

A1

3

(b) 2*m* – 2 × 3 = *m* × 0.5 + 3 × 0.5

Four term equation for conservation of momentum with ±0.5 for both velocities (no marks for 3m × 0.5)

M1

Correct equation

A1

1.5*m* = 7.5
m = 5 kg

Correct mass for velocity used

A1

or

$2m - 2 \times 3 = m \times (-0.5) + 3 \times (-0.5)$

Equation for conservation of momentum with opposite sign for the 0.5

M1

$2.5m = 4.5$

$m = 1.8$ kg

Correct mass for the velocity used

A1

5

[8]

Q23.

(a) C.L.M.

$m(4\mathbf{i} + 3\mathbf{j}) + 2m(-\mathbf{2i} + 2\mathbf{j}) = mv + 2m(\mathbf{i} + \mathbf{j})$

M1

$7\mathbf{j} = v + (2\mathbf{i} + 2\mathbf{j})$

$v = -2\mathbf{i} + 5\mathbf{j}$

A1 for one slip

A2,1,0

3

(b) The angle with **j** direction:

*OE. in **i** direction*

A: $\tan^{-1}\frac{2}{5} = 21.8°$

B: $\tan^{-1}\frac{1}{1} = 45°$

M1 for two inverse tan and addition of angles

M1

The angle = 21.8° + 45° = 67°

AWRT.

Alternative (not in the specification)

$(-2\mathbf{i} + 5\mathbf{j}).(\mathbf{i} + \mathbf{j}) = \sqrt{29} \times \sqrt{2} \cos\theta$ *(M1)*

$\cos\theta = \frac{3}{\sqrt{58}}$ *(A1)*

$\theta = 67°$ *(A1F) awrt*

(c) The impulse = Gain in momentum of A

M1

$= m(-2\mathbf{i} + 5\mathbf{j}) - m(4\mathbf{i} + \mathbf{3j})$

A1F

$= -6m\mathbf{i} + 2m\mathbf{j}$

A1F

3

(d) $-3\mathbf{i} + \mathbf{j}$ or any scalar multiple of $-3\mathbf{i} + \mathbf{j}$

B1

1

[10]

Q24.

(a) $3 \times 4 \times 2 \times (-4) = 5v$

Three term equation for conservation of momentum. Correct equation

M1A1

$4 = 5v$

$v = \frac{4}{5} = 0.8$

Correct speed (for use of mg instead of m deduct the first A1)

A1

s3

(b) $3 \times 4 \times 2 \times (-4) = 3 \times 0.4 + 2v$

Four term equation for conservation of momentum. Correct equation

M1A1

$4 = 1.2 + 2v$

$v = \frac{4 - 1.2}{2} = 1.4$

Correct speed
(for use of mg instead of m deduct the first A1)

A1

3

[6]

Q25.

(a) $2\begin{bmatrix}3\\-2\end{bmatrix}+3\begin{bmatrix}-4\\1\end{bmatrix}=5\mathbf{v}$

Three term vector equation, with a '+' sign, for conservation of momentum

M1

Correct equation
Deduct this first A mark for use of mg

A1

$\mathbf{v}=\frac{1}{5}\begin{bmatrix}-6\\-1\end{bmatrix}=\begin{bmatrix}-1.2\\-0.2\end{bmatrix}$

Correct velocity

A1

3

(b) $v=\sqrt{1.2^2+0.2^2}$ = 1.22 m s^{-1}

Finding speed from their velocity in part (a) (Must include addition of two terms)

M1

Correct speed from their velocity
Accept 1.21

A1F

2

[5]

Q26.

(a) Conservation of momentum:
$0.3(3) - 0.2(2) = 0.3v_A + 0.2v_B$

M1A1

$3v_A + 2v_B = 5$ ----------------(1)
Newton's experimental law :

$0.8 = \frac{v_B - v_A}{5}$

M1

$v_B - v_A = 4$ -----------------(2)

For both (1) and (2)

A1

Solving (1) and (2)

Dependent on both M1s

m1

$v_B = 3.4$
$v_A = -0.6$

For both solutions

A1F

6

(b) $0.7 = \dfrac{v}{3.4}$

M1

$v = 2.38$

A1F

Speed of B (2.38) > Speed of A (0.6)
$\therefore$ B collides again with A

Cannot be gained without A1F

E1

3

[9]

Q27.

(a) $\rightarrow$ 6ms^{-1}

2 kg ○ ○ 3 kg

$\rightarrow v$

$2 \times 6 = 3 \times v$

M1

$v = 4$ ms^{-1}

A1

A1

3

(b) $\rightarrow$ 6ms^{-1}

2 kg ○ ○ 3 kg

$\leftarrow v$ $\rightarrow 4v$

$2 \times 6 = -2 \times v + 3 \times 4v$

M1 all terms

M1A1

$12 = 10v$

$v = 1.2$ ms^{-1}

ft sign error (v = 0.857)

A1ft

3

[6]

Q28.

(a) $m(5\mathbf{i} - 3\mathbf{j}) + 0.2(2\mathbf{i} + 3\mathbf{j})$

Momentum terms added

M1

All correct

A1

2

(b) (i) $(0.2 + m)(k\mathbf{i} + \mathbf{j})$

Seen, or used to find m

B1

use of conservation of momentum

Used with candidate's expressions in 2D equation or used to give one of the 1D equations below

M1

$-3m + 0.6 = 0.2 + m$

$m = 0.1$

Full verification accepted CAO

A1

3

(ii) $5m + 0.4 = 0.2k + mk$

A1

substitute *m*

m1

$k = 3$

A1

3

[8]

Q29.

(a) conservation of momentum

$mu = mv_A + mv_B$

M1

$u = v_A + v_B$

A1

restitution

$eu = v_B - v_A$

OE

M1A1

$v_B = \frac{1}{2}u(1+e)$

OE

A1F

5

(b) $mv_B = mw_B + 2m\frac{3u}{8}$

M1A1

$ev_B = \frac{3u}{8} - w_B$

OE

M1A1

Elimination of w_B

dependent on both M1s

m1

$4e^2 + 8e - 5 = 0$

simplified quadratic equation in e only

A1F

$e = \frac{1}{2}$

stated as the only value (0 < e < 1 for follow through)

A1F

7

[12]

Q30.

(a) $12\begin{bmatrix}4\\7\end{bmatrix}+4\begin{bmatrix}2\\3\end{bmatrix}=16\boldsymbol{v}$

Three term momentum equation

M1

Correct equation

A1

$\boldsymbol{v}=\frac{1}{16}\begin{bmatrix}56\\96\end{bmatrix}=\begin{bmatrix}3.5\\6.0\end{bmatrix}\text{ms}^{-1}$

*Solving for **v***

m1

Correct velocity

A1

4

(b) $12\begin{bmatrix}4\\7\end{bmatrix}+4\mathbf{u}=16\begin{bmatrix}1\\4\end{bmatrix}$

Three term momentum equation

M1

Correct equation

A1

$\mathbf{u}=\frac{1}{4}\begin{bmatrix}-32\\-20\end{bmatrix}=\begin{bmatrix}-8\\-5\end{bmatrix}\text{ms}^{-1}$

Correct velocity

A1

3

[7]

Q31.

(a) $8 \times 8 + 12 \times 6 = 20v$

M1:Three term equation for conservation of momentum

M1

$v = \frac{136}{20} = 6.8 \text{ ms}^{-1}$

A1: Correct equation

A1

A1: Correct v

A1

3

(b) $8 \times 8 + 12 \times 6 = 8 \times 6.5 + 12v$

M1: Four term conservation of momentum equation

M1

$v = \frac{84}{12} = 7$ ms^{-1}

A1: Correct equation

A1

A1: Correct v

A1

3

[6]

Q32.

(a) $m\begin{bmatrix}4\\2\end{bmatrix} + 3\begin{bmatrix}-1\\-1\end{bmatrix} = (m+3)\begin{bmatrix}1\\V\end{bmatrix}$

M1: Conservation of momentum equation with 3 terms

M1

$4m - 3 = m + 3$

A1: Correct momentum equation

A1

$3m = 6$

M1: Solving equation

M1

$m = 2$ **AG**

A1: Correct m from correct working
Note: Deduct one mark for using mg instead of m.

A1

4

(b) $4 - 3 = 5V$

M1:Conservation of momentum equation for component containing V

M1

A1: Correct equation

A1

$V = 0.2$

A1: Correct V

A1

3

[7]

Q33.

(a) By P.C.L.M.:

$4mu = 4mv + 2\,mv$

M1

$2u = 2v + v$ …………(1)

Law of restitution: $e = \dfrac{v_2 - v_1}{u}$ ……(2)

A1 for both correct

M1A1

Solving (1) and (2) →

Dependent on both Ms

m1

$v = \dfrac{2u - eu}{3}$

$v_2 = \dfrac{2u + 2eu}{3}$

A1F for both v_1 and v_2

A1F

5

(b) $\dfrac{12mu}{5} = 4mu - 4m\left(\dfrac{2u - eu}{3}\right)$

Impulse / Momentum

M1A1F

$\left[\text{or} = 2m\left(\dfrac{2u + 2eu}{3}\right)\right]$

$20meu = 16\,mu$

$e = \dfrac{4}{5}$

Solution to get the right answer
Answer given

A1

3

(c) $2mv = 2mv + mv$

M1

$$\frac{v_4 - v_3}{v_2} = \frac{4}{5}$$

M1

$$v = \frac{12u}{25}$$

m1

Dependent on both Ms

A1F

4

(d) $v = \dfrac{2u - \frac{4}{5}u}{3} = \dfrac{2u}{5}$

M1

$v \succ v \Rightarrow A$ and B will not collide again

E1F

2

[14]

Q34.

(a) $0.1 \begin{bmatrix} 8 \\ 12 \end{bmatrix} + 0.2\text{V} = 0.1 \begin{bmatrix} -2 \\ 6 \end{bmatrix} + 0.2 \begin{bmatrix} 6 \\ 0 \end{bmatrix}$

M1: correct use of momentum principle and 4 momentum terms

M1A1

$0.2\,\mathbf{V} = \begin{vmatrix} 0.2 \\ 0.6 \end{vmatrix}$

ft one slip

A1F

$\mathbf{V} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

ft one slip

A1F

4

(b) (i) $S_A = \begin{bmatrix} -4 \\ 12 \end{bmatrix}$ $S_B = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$

B1B1

2

(ii) $\mathbf{d} = \pm \begin{bmatrix} 16 \\ -12 \end{bmatrix}$

Attempt at subtraction

M1

$|\mathbf{d}| = \sqrt{\left(16^2 + (-12)^2\right)} = 20$

M1: magnitude of vector with two non-zero terms, + needed

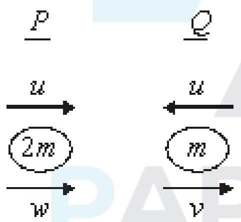
M1A1F

3

[9]

Q35.

(a) (i)



Conservation of momentum

$2mu - mu = mv + 2mw$

M1A1

M1 one momentum term correct

$u = v + 2w$ (1)

Restitution

$v - w = 2ue$ (2)

M1 e × speed of approach seen

M1A1

(1) – (2) gives $3w = u - 2ue$

M1

$w = \frac{u}{3}(1 - 2e)$

A1

(1) + 2(2) gives u (1 + 4e) = 3v

$v = \frac{u}{3}(1 + 4e)$

B1ft

7

(ii) v always positive, so same direction when $\frac{u}{3}(1 - 2e) > 0$

For > 0 or solving = 0

M1

$\therefore$ 1 – 2e > 0

Must be convincing about <

A1

$e < \frac{1}{2}$

2

(b) (i) $I = mv - mu$

Use of mv – mu

M1

$= m\frac{u}{3}(1 + 4e) + mu$

Paired speeds correct

A1

$= 4\frac{mu}{3}(1 + e)$

Printed answer

A1

3

(ii) $0 \le e < \frac{1}{2}$

Use of e values in I

M1

$\therefore \frac{4mu}{3}(1 + 0) \le I < \frac{4mu}{3}(1 + \frac{1}{2})$

$\frac{4mu}{3} \le I < 2mu$

Printed answer

A1

2

[14]

Q36.

$12m = 3mv + 3mv$

M1 Momentum terms, all present, accept one slip

M1A1

$v = 2$

ft slip

A1F

[3]

Q37.

(a)

impulse = $mv - mu$

attempt to use $|mv - mu|$

M1

= 240(1) – 240(–2)

correct signs

A1

= 720 N s

must have units; ft applies to 240 N s only

A1ft

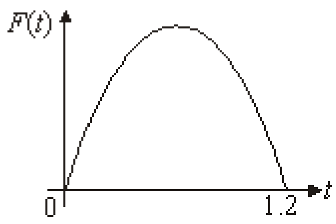
3

(b) (i) t = 0, 1.2

B1

1

(ii)



shape

B1

symmetrical / axis / labels

B1

2

(iii) max when $t = 0.6$ (symmetry)

t value found/stated

B1

F (0.6) = 500(0.6)(6 – 5 × 0.6)

attempt to find F

M1

= 300 × 3
= 900 N

AG

A1

3

(iv) total area below curve = impulse magnitude

E1

1

[10]

Q38.

(a) Impulse = area

M1

= 2000 × 5 + $\frac{1}{2}$ (2000) × 5

Rectangle or triangle or trapezium area

M1

= 15 000 N s

AG

A1

3

(b) |Change in momentum| = impulse

Attempt at |mv – mu| = impulse from (a)

M1

1000 v – 0 = 15000

$v = 15 \text{ ms}^{-1}$

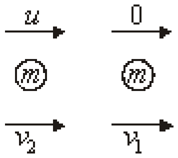
Must state units

A1

2

[5]

Q39.



(a) Restitution: $v_1 - v_2 = eu$

Attempt at restitution

M1

Momentum: $mv_1 + mv_2 = mu$

Attempt at momentum

M1

$v_1 - v_2 = eu$ (1)

$v_1 + v_2 = eu$ (2)

Both correct

A1

(1)+(2) $2v_1 = u(1 + e)$

M1

$v_1 = \frac{u}{2}(1 - e)$

AG

A1

(2)–(1) $2v_2 = u(1 - e)$

A1

$v_2 = \frac{u}{2}(1 - e)$

B1f.t.

6

(b) Speed = $\frac{2}{3} \times \frac{u}{2}(1 + e) = \frac{u}{3}(1 + e)$

B1 **1**

(c) (i) $\frac{u}{3}(1+e) = \frac{u}{2}(1-e)$

Equating

M1

2 + 2e = 3 – 3e
5e = 1
$e = \frac{1}{5}$

Solving or showing

A1 **2**

(ii) Speed = $\frac{2u}{5}$

B1 **1**

(d) B reached wall after $\frac{5d}{3u}$

Attempt to find twice

M1A1f.t.

In this time A travels $\frac{5d}{3u} \times \frac{2u}{5} = \frac{2d}{3}$

Attempt to find distance

M1A1f.t.

A and B now have same speed so meet at

half remaining distance = $\frac{1}{2}\left(\frac{d}{3}\right)$

Attempt to find remaining distances

M1

= $\frac{d}{6}$

Special case $\frac{5d}{6} \Rightarrow$ *5 marks*

A1 **6**

Alternative:
Ratio of speeds after collision = 1.5 : 1

(M1A1f.t.)

Ratio of distance after collision = 1 : $\frac{2}{3}$

(M1A1f.t.)

Then $d - \frac{2}{3}d$ left = $\frac{d}{3}$

(M1)

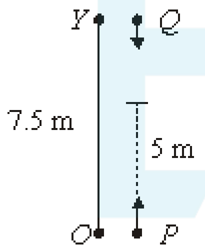
Same speed to meet half way = $\frac{1}{2}\left(\frac{d}{3}\right) = \frac{d}{6}$

(A1)

[16]

Q40.

(a) (i)



Q : $s = ut + \frac{1}{2}at^2$

$s = 0 + \frac{1}{2}\ 9.8 \times \frac{5}{7}$

M1 : full method for s

M1

$s = 2.5$

A1

2

(ii) 5 + 2.5 = 7.5 so collision occurs

A1

1

(b) Q: $v = 0 + 9.8 \times \frac{5}{7}$

M1 : full method for v

M1

$= 7$ A1

$\downarrow - 0.2 \times 3.5 + 0.3 \times 7 = 0.5v$

M1: Momentum equation with 3 terms with appropriate masses.
Ft velocity of Q

M1A1F

$v = 2.8\downarrow$

A1F for magnitude
f.t. one minor slip in working

A1F

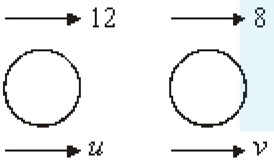
6

A1 for direction, (may be implied in answer given in vector from with negative component)
SC B1 for v = – 2.8

A1

[9]

Q41.



Cons of momentum,
$12\,(800) + 8\,(1000) = 800\,u + 1000\,v$

Attempt at cons of mom – 2terms correct
Fully correct – accept if 10 used

M1A1

$88 = 4u + 5v \quad —(1)$ Restitution law,

$v - u = \frac{1}{8}(12 - 8)$

Attempt at restitution equations
Fully correct – accept if 10 used

M1 A1

$v - u = \frac{1}{2} \quad —(2)$

$4 \times (2) + (1)$ gives $9v = 90$

Solving a pair of simultaneous eqns

M1

v = 10 ms^{-1}

A1

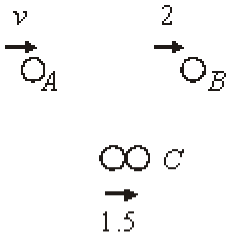
∴ u = 9.5 ms^{-1}

AG; v = 10 correctly obtained from pair of eqns

B1

[7]

Q42.



(a) $3mv + 2m = 4m \times 1.5$

M1A1

$v = \frac{4}{3}$

A1

3

(b) $v^2 = u^2 + 2as$

$0 = (1.5)^2 + 2 \times a \times 3$

Full method, accept ±

M1A1

retardation =0.375 m s^{-2}

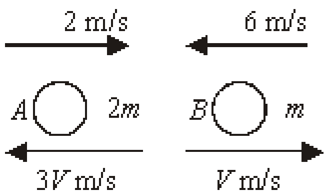
Magnitude required

A1

3

[6]

Q43.



Conservation of momentum:

$4m - 6m = -6mV + mV$

M1 at least 3 momentum terms; A1 each side

M1A1A1

$V = \frac{2}{5}$

*ft one or two errors in **4** term equation*

A1F

[4]

Q44.

(a) $A \xrightarrow{43.2}$ $B \xrightarrow{0}$

$1200 \xrightarrow{v}$ $1500 \xrightarrow{u}$

Conservation of momentum

$43.2(1200) = 1200v + 1500u$

Use of km h^{-1} or ms^{-1} throughout

M1A1

$43.2 = v + 1.25u$

(1)

Restitution Law

$u - v = \frac{1}{4}(43.2 - 0)$

M0 if 'e' on the wrong side

M1A1

$10.8 = u - v$

(2)

(1) + (2) $54 = 2.25u$

Solving a part of equations

M1

$u = 24$ km h^{-1}

or $6\frac{2}{3}$ *ms^{-1}*

A1

$v = 13.2$ km h^{-1}

Follow through error through error in their *equations or* $3\frac{2}{3}$ *ms*$^{-1}$

B1f.t.

7

(b) 24 km h^{-1} = $\frac{24 \times 1000}{60 \times 60}$

= $6\frac{2}{3}$ ms^{-1}

Conversion of units correct in this part

B1

Impulse = 1500$\left(6\frac{2}{3}\right)$ – 0

Impulse = change in momentum for the particle

M1

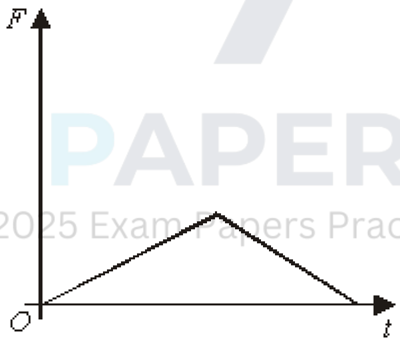
= 10 000 N s

Printed answer

A1

3

(c) (i)



B1 increase then decrease
B1 all correct

B(2,1,0)

2

(ii) Area $\Rightarrow \frac{1}{2}(0.01)F = 10000$

Use of area – uses printed answer

M1A1

$F = 2 \times 10^6$ N

Follow through error in area.

A1f.t.

3

(iii) Likely to be non-linear, or force wouldn't change abruptly

Suitable comment – dependent on graph

E1

1

[16]

Q45.

(a) $\mathbf{v} = \left(\begin{bmatrix} 2 \\ 8 \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \end{bmatrix}\right) \div 3$

All work in this question must be with component of vectors, which must be recombined into columns or **i**, **j** *form for final accuracy marks.*

Difference of position vectors

M1

Division by 3, dependent on first M1

m1

$= \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

CAO

Alternative:

Integration of $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$ *wrt time or* $3(\mathbf{i} + 2\mathbf{j}) \Rightarrow$ *M1;*
Use of one set of values of t and **r** *(to find constant), giving*
$\mathbf{r} = (1 - 1)\mathbf{i} + (2t + 2)\mathbf{j} \Rightarrow$ *A1; verify* 2^{nd} *point* $\Rightarrow$ *A1*

A1

3

(b) $0.2\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0.1\begin{bmatrix} -5 \\ 5 \end{bmatrix}$

sum *of momentum terms i.e. m**v**, max 1 miscopy in* ***v***
or m (masses swapped = one miscopy):
M0 A0 if g included
Ratio of masses M1 A0 M1 A1

M1

$= \begin{bmatrix} -0.3 \\ 0.9 \end{bmatrix}$

ft (a)

A1F

$= 0.3\mathbf{v}$

Conservation of momentum; may have lost 1^{st} *M1*

SC: if g included M0 A0 M1 A1

$\mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

ft (a) or one slip

A1F

4

(c) $3 \times \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

M1

$= \begin{bmatrix} -3 \\ 9 \end{bmatrix}$

ft from (b)

A1F

$+ \begin{bmatrix} 2 \\ 8 \end{bmatrix}$

m1

$= \begin{bmatrix} -1 \\ 17 \end{bmatrix}$

ft from (b) or one slip

A1F

4

Alternative:

$\int \begin{bmatrix} -1 \\ 3 \end{bmatrix} \mathrm{d}t$ *attempted*

M1

$\begin{bmatrix} -t \\ 3t \end{bmatrix}$

A1F

t = 0, r = $\begin{bmatrix} 2 \\ 8 \end{bmatrix}$ *used*

m1

$t = 3 \Rightarrow \begin{bmatrix} -3 \\ 9 \end{bmatrix} + \begin{bmatrix} 2 \\ 8 \end{bmatrix} = \begin{bmatrix} -1 \\ 17 \end{bmatrix}$

A1F

[11]

Q46.

(a) (i) $\frac{1}{4}\begin{pmatrix}3\\6\end{pmatrix} - \frac{1}{4}\begin{pmatrix}-1\\4\end{pmatrix}$

Difference of vectors for M1

M1

$= \begin{pmatrix}1\\0.5\end{pmatrix}$ kg m s^{-1}

Answer

A1

Units

A1

3

(ii) $\begin{pmatrix}1\\0.5\end{pmatrix}$ kg m s^{-1}

SC Accept correct answer if (a)(i) wrong
ft vector answers

B1F

1

(b) $0.1\begin{pmatrix}5\\2\end{pmatrix} - 0.1(v) = \begin{pmatrix}1\\0.5\end{pmatrix}$

Difference of two momentum vectors attempted for M1

M1A1

$v = \begin{pmatrix}-5\\-3\end{pmatrix}$ m s^{-1}

Last mark for division by 0.1

A1F

3

[7]

Q47.

(a) $8 \times 5 + 4 \times (-5) = 4v$

Three term momentum equation.

M1

Correct LHS.

A1

Correct RHS.

A1

$v = \frac{20}{4} = 5\ \text{ms}^{-1}$

Correct speed (accept ± 5)

A1

4

(b) $20 = 4 \times 2v + 8v$

Three term momentum equation with $v_B = 2v_A$

M1

Correct equation.

A1

$v = \frac{20}{16} = 1.25$

Correct v_A

A1

$v = 2.5\ \text{ms}^{-1}$

Correct v_B

A1

4

[8]

Q48.

(a) C of momentum

$m.63u = 81m.v$

M1A1

$v = \frac{7}{9}u$

A1

3

(b) If x is the number of arrows required,

$x\ m.63u = (80m + xm)7u$

M1A1

$9x = 80 + x$

M1

$x = 10$

Needs unknown on both sides

A1

4

[7]

Q49.

(a) m $4m$

Initial $\rightarrow$ $2u$

Final $\rightarrow v$ $\rightarrow$ v

C of momentum:
$4m.2u = (m + 4m).\ v$

M1 A1

$8m.u = 5mv$

$v = \frac{8}{5}u$

A1

3

(b) Impulse = change in momentum
Using particle P:

Impulse = m × $\frac{8}{5}$u

M1

$= \frac{8}{5}mu$

A1

2

[5]

Q50.

(a) $3 \times 5 + 2 \times (-5) = 5v$

Three term conservation of momentum equation

M1

Correct equation

A1

$v = \frac{5}{5} = 1\text{ms}^{-1}$

Correct velocity

A1 3

(b) $3 \times 5 + 2 \times (-5) = 2v + 3 \times 0.5$

Four term conservation of momentum equation

M1

Correct equation

A1

Solving for velocity

m1

$v = \frac{3.5}{2} = 1.75 \text{ ms}^{-1}$

Correct velocity

A1 4

[7]

Q51.

(a) Initial $\rightarrow 3u$ $\leftarrow u$

$6m$ $2m$

Final $\rightarrow V$ $\rightarrow V$

C of momentum

$6m.3u - 2m.u = 6m.V + 2m.V$

M1

$16u = 6V + 2V$

A1

Restitution $\frac{1}{4}(3u + u) = V - V$

M1A1

$u = V - V$

$14u = 8V$

M1

$V = \frac{7}{4}u$

A1 6

(b)

$V = u + \frac{7}{4}u$

M1

$= \frac{11}{4}u$

A1

2

[8]

Q52.

(a) $0.1 \times 5 + 0.4 \times 3 = 0.5v$

Using conservation of momentum

Correct equation

M1 A1

$v = \frac{1.7}{0.5} = 3.4\ \text{ms}^{-1}$

Correct v

A1

3

(b) $0.1 \times 5 + 0.4 \times 3 = 0.1v + 0.4 \times 3.5$

Using conservation of momentum

Correct equation

M1 A1

$v = \frac{1.7 - 1.4}{0.1} = 3\ \text{ms}^{-1}$

Correct v

m1 A1

4

[7]

Q53.

(a) Using conservation of momentum

M1

$3m\begin{pmatrix} 7 \\ -8 \end{pmatrix} + m\begin{pmatrix} 2 \\ 5 \end{pmatrix} = m\begin{pmatrix} 5 \\ -4 \end{pmatrix} + 3m\mathbf{v}$

A1

$\begin{pmatrix} 21 \\ -24 \end{pmatrix} + \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \end{pmatrix} + 3\mathbf{v}$

M1

$3\mathbf{v} = \begin{pmatrix} 18 \\ -15 \end{pmatrix}$

$\mathbf{v} = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$

A1

4

(b) Change in momentum =

$m\begin{pmatrix} 5 \\ -4 \end{pmatrix} - m\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

M1 for $-3m\mathbf{i} + 9m\mathbf{j}$

M1

$= 3m\mathbf{i} - 9m\mathbf{j}$

sc *1 for* $3\mathbf{i} - 9\mathbf{j}$

A1

2

[6]

Q54.

(a) $2 \times 4 = 2 \times 1 + 4v$

Three term equation for conservation of momentum, with $u_B = 0$

M1

Correct equation

A1

$v = \dfrac{8-2}{4} = 1.5$

Correct velocity
(use of mg deduct 1 mark)

A1

3

(b) $4 \times 1.5 = 4v + m \times 2$

Three term equation for conservation of momentum, with $u_C = 0$

M1

Correct equation

A1ft

$v = \frac{6-2m}{4}$

Correct velocity

A1ft

3

(c) $1 > \frac{6-2m}{4}$

Equation or inequality with v from previous answer and 1

M1

Correct inequality

A1ft

$4 > 6 - 2m$

Solving for m

m1

$2m > 2$

$m > 1$

ag *Correct result from correct working*

A1

4

[10]

Q55.

(a) Velocity of Q is $2u$

Impulse = $3m.2u$

M1

= $6mu$

A1

2

(b) (i) Initial $\rightarrow 2u$

Q $3m$ R $5m$

Final $\rightarrow V$

C of momentum $3m.2u = 5mV$

M1A1

$V = \frac{6}{5}u$

A1

3

(ii) Restitution $e.2u = V$

M1A1

$\therefore\ e = \frac{3}{5}$

A1

2

(iii) Before P collides with Q again,
P travels distance a at speed $2u$

M1

$\therefore$ Time = $\frac{a}{2u}$

A1

2

(c) After impact velocity of B is V
Conservation of momentum $mu = MV$

B1

Restitution $eu = V$

$eMu = MV = mu$
$e \leq 1$

M1

$m \leq M$

A1

3

[12]

Q56.

(a) $2 \times 5 = 40v$

Three term conservation of momentum equation

M1

$v = \frac{10}{40} = 0.25\ \text{ms}^{-1}$

Correct v

A1

2

(b) $2 \times 6 + 40 \times 0.25 = 42v$

Three term conservation of momentum equation

M1

Correct equation

A1

$v = \frac{22}{42} = \frac{11}{21} = 0.524 \text{ ms}^{-1}$ (to 3sf)

Correct v

A1

3

[5]

Q57.

(a) $3 \times 3 + 2 \times 2 = 5v$

Conservation of momentum equation

M1

$v = \frac{13}{5} = 2.6 \text{ ms}^{-1}$

A1

2

(b) $5 \times 2.6 - 5 \times 1 = 10v$

Three term equation for conservation of momentum

Correct equation

M1

$v = \frac{8}{10} = 0.8 \text{ ms}^{-1}$

Correct v

A1

3

(c) $0 = 0.8 + 0.2a$

Constant acceleration equation with $v = 0$

M1

$a = -4 \text{ ms}^{-2}$

Correct acceleration

A1 ft.

F = 10 × 4 = 40N

Correct force

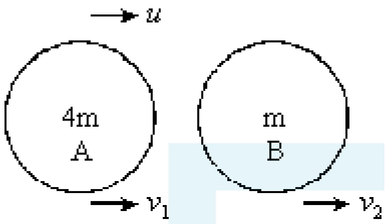
A1 ft

3

[8]

Q58.

(a) Initial



C of Momentum: $4mu = 4mv_1 + mv_2$

M1A1

$v_2 = 2v_1$

$\Rightarrow 4u = 4v_1 + 2v_1$

M1

$\therefore v_1 = \frac{2}{3}u$

A1

4

(b) Restitution: $e\,u = v_2 - v_1$

M1A1

$v_2 = \frac{4}{3}u$

B1

$\therefore e\,u = \frac{4}{3}u - \frac{2}{3}u$

M1

$e = \frac{2}{3}$

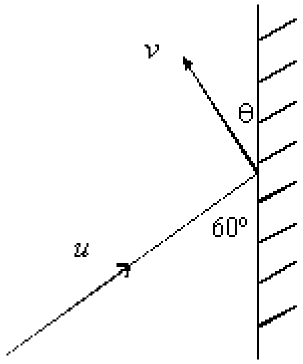
A1

5

[9]

Q59.

(a)



Velocity // wall unaltered

M1

$u \cos 60 = v \cos \theta$

A1

Velocity perp to wall

$e\, u \sin 60 = v \sin \theta$

M1A1

Dividing: $\tan \theta = e \tan 60$

$= \frac{3}{4}\sqrt{3}$

$\therefore\ \theta = 52.4°$

A1

$\therefore$ Direction of motion is changed by 112.4°

A1

6

(b) Impulse is change in momentum perp to wall

M1

$= mu \sin 60 + mv \sin \theta$

M1

$= mu \sin 60\,(1 + e)$

A1

$= 0.3 \times 5 \times \frac{\sqrt{3}}{2} \times 1.75$

$= \frac{21}{16}\sqrt{3}$

A1

Time × Impulse = change in momentum

M1

$\therefore$ Impulse = 20 × $\frac{21\sqrt{3}}{16}$

A1

= 45.5

A1

7

[13]

Q60.

(a) Impulse = change in momentum

M1

$J = 2m \times 3\mu$

$J = 6mu$

A1

2

(b) Conservation of momentum

M1

$6mu = 2mv_a + mv_b$

A1

Restitution

M1

$3u \times e = 3u \times \frac{3}{5} = v_a - v_a$

A1

$6u = 2v_a + v_b$

Subtracting $\frac{21}{5}u = 3v_a$

$\frac{7}{5}u = v_a$

A1

$6u + \frac{18}{5}u = 3v_b$

$v_b = \frac{16}{5}u$

A1

6

(c) Velocity after impact is e × velocity before impact

M1

$= \frac{1}{2} \times \frac{16}{5}u$

$= \frac{8}{5}u$

ft [from (b)]

A1ft

2

(d) Time taken for B to strike wall after collision is $\frac{4a}{\frac{16u}{5}}$

In this time A has travelled $\frac{7}{5}u \times \frac{5a}{4u}$

$= \frac{7}{4}a$

B1

Two spheres are $\frac{9}{4a}$ apart

ft [from above]

B1ft

Speeds of spheres are (A) $\frac{7}{5}u$ and (B) $\frac{8}{5}u$

A1

Distance from the wall is $\frac{8}{15} \times \frac{8}{4}a$

M1

$= \frac{6}{5}a$

A1

7

[17]

Q61.

(a) Using conservation of momentum

M1

$mu = (m + 3m + km)v$

M1

$\therefore \ v = \frac{u}{4 + k}$

A1

3

Or

First jerk; conservation of momentum

$mu = (m + 3m)V$

$V = \frac{1}{4}u$

B1

Second jerk;

$(m + 3m)V = (m + 3m + km)v$

M1

$v = \frac{u}{4 + k}$

A1

(b) Velocity of A and B after first string

becomes taut is $\frac{1}{4}u$

Could have been seen in (a)

B1

When second string becomes taut;

I_{AB} is impulse in string AB,

I_{BC} in string BC

Using impulse = change in momentum,

Could be for I_{BC}

M1

Sphere A, $I_{AB} = m\left(\frac{1}{4}u - \frac{u}{4+k}\right)$

M1

$= \frac{mku}{4(4+k)}$

A1

Sphere C, $I_{BC} = mk \cdot \frac{u}{4+k}$

$= \frac{mku}{4+k}$

From (a)

A1ft

Ratio is $\frac{1}{4}$: 1

Or 1 : 4

Which is independent of k

sc 5 if given impulses acting on A, B *and* C

A1

6

[9]

Q62.

(a) $20 \times 0.5 = 10.5v$

Conservation of momentum equation with two non-zero terms

M1

$v = \frac{10}{10.5} = \frac{20}{21}$ ms^{-1}

Correct answer from correct working

A1

2

(b) $20 \times 0.5 + 10.5 \times \frac{20}{21} = 11v$

Conservation of momentum equation with three non-zero terms

M1

$v = \frac{20}{11} = 0.82$ ms–1

Correct equation
Solving for v
Correct answer

A1/M1/A1

4

[6]

Examiner reports

Q1.

This question proved to be a successful starter with 68% of students choosing the correct answer.

The incorrect answer that was most often chosen was $\begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix}$, indicating that students had forgotten to divide by the combined mass.

Q2.

Parts of this question proved to be the most challenging on the paper, with mean marks of 59%, 47%, 13% and 32% for (a), (b), (c) and (d) respectively.

The concepts of conservation of momentum and the law of restitution were clearly familiar concepts but students made errors through lack of useful, clearly labelled diagrams. In the best solutions, students drew a diagram and labelled velocities clearly. Where labelling was not clear, it became apparent that students had confused themselves, and their momentum and restitution equations were not consistent with directions. Many students were unable to justify the printed answer for speed if they had obtained an answer of $\frac{u(2e-3)}{5}$ first.

Almost all students were able to score at least 1 mark in (b), with the most common error being in substituting the expression for speed rather than velocity.

Part (c) required an appreciation of the range of values for e in order to justify the directions of motion of the two smooth spheres. Many students failed to offer any justification and scored zero marks.

For part (d) many students were able to quote the relevant formula for impulse and to substitute corresponding expressions for velocities, but sign errors limited the number of marks available. The final result could then be obtained by considering limiting values for e. A significant number of students did not attempt this part at all.

Q4.

Some students did very well on this question, but there were two main stumbling blocks. The first was for those students who did not appreciate the significance of the fact that the two particles were moving towards each other. This resulted in them treating the particles as if they were both moving in the same direction thus finding the total momentum before the collision to be 20 + 12 = 32 rather than 20 – 12 = 8. The second issue was not introducing a negative sign with the 0.6 when considering the velocity in the second case. Some students thought that the second speed could be found by considering the particles as if they coalesced during the collision.

Q5.

This question was done very well by the vast majority of students with many correct solutions. Some students worked in grams instead of kilograms. The most common error was to include a 200 in the calculation for the total momentum before the collision.

Q6.

(a) The principle of conservation of linear momentum and the law of restitution were well applied by a great majority of students. However some students lost one or both of the last two accuracy marks by not simplifying their expressions for v_A and v_B.

(b) Here it was required to state that as e $\leq$ 1, then $v_B \leq \frac{u}{2}(1 + 5)$, or equivalent, and hence $v_B \leq 3u$. Many students gave satisfactory answers, but some lost marks due to either using a strict inequality or by not giving a fully convincing answer.

(c) Many students lost marks here for using the speed of B together with the mass of A to state the difference of the two momentums.

Q7.

This question was done very well, with almost all candidates gaining full marks. The two main areas where marks were lost were simple arithmetic errors and candidates not using the total mass of 10 kg in their equations.

Q8.

This question was done very well, with many candidates gaining full marks. The candidates who did not progress on this question had difficulties setting up the equation for the conservation of momentum.

Q9.

This question was done very well, with many candidates gaining full marks. The candidates who did not progress on this question had difficulties setting up the equation for the conservation of momentum.

Q10.

This question was generally done very well. The most common error was to use $5m$ instead of $(5 + m)$ for the mass of the trolleys after the collision.

Q11.

There were many correct solutions to this question. There were some candidates who were unable to form the required equations, although a few could write down a correct vector equation but could not extract equations from it to find the two unknowns. There were also candidates who wrote down correct equations and made algebraic or arithmetic errors when solving them. In addition, there were a few errors due to incorrect signs.

Q12.

Almost all of the candidates produced correct solutions and gained full marks.

Q13.

While there were many good solutions to this question, some candidates made little progress. Some candidates made errors when trying to write down a vector equation based on conservation of momentum. The most common of these errors was to use a mixture of scalars and vectors in the equation. Some candidates were able to form a correct vector equation, but had difficulty extracting the correct equations for each component.

Q14.

This question was attempted well, with many candidates scoring full marks. The candidates were able to use the principle of conservation of linear momentum and Newton's experimental law correctly to answer part (a). However, in part (b), a small number of candidates seemingly forgot to find the speed of B immediately after the collision. These candidates lost the last method and accuracy marks.

Part (c) proved to be too challenging for a significant number of candidates. These candidates attempted to answer this part by substituting values such as 9 or 10 for x. The more able candidates stated that $vB < 0$ for a further collision, and proceeded by solving an inequality to show the required result.

Q15.

This question proved challenging for a large number of candidates.

Many candidates were unable to provide a valid reason as to why the components of the velocities of A and B parallel to the unit vector **j** were unchanged by the collision. The response here was often "because **j** is perpendicular to **i**" or "because the spheres collide horizontally".

In part (b), a lot of candidates did not understand that the coefficient of restitution is the ratio of two speeds and not velocities. There were candidates who stated in part (a) that the components of the velocities parallel to the unit vector **j** are not changed, but then effectively ignored their own assertion when attempting to answer part (b).

Q16.

This question was done well, with almost all of the candidates gaining full marks. The errors that were seen were mainly due to taking the velocities as positive or making simple addition errors.

Q17.

This question was done well, with almost all of the candidates gaining full marks. The errors that were seen were mainly due to taking the velocities as positive or making simple addition errors.

Q18.

While many candidates gained full marks on this question, there were two common reasons for losing marks. The first, in part (a), was simply arithmetic errors, which seemed to be quite common in this question. The second, also in part (a), was candidates' not including the combined mass of 10 kg in their working. This often led to answers such as $\begin{bmatrix} 11 \\ 22 \end{bmatrix}$. In part (b), many candidates who had made errors were able to gain the method marks.

Q19.

Many candidates were able to use the principle of the conservation of linear momentum and Newton's experimental law correctly to answer part (a) of this question. Many candidates were able to find the velocity of the sphere A. However, not all of these candidates were able to use the given fact that the direction of motion of the sphere A is reversed to form an inequality to answer part (b).

Even fewer candidates answered part (c) correctly. The challenge in answering the last part of the question was getting the signs of the velocities of the spheres correct as well as the direction of the inequality correct.

Q20.

There were many good responses to this question. Parts (a) and (b) were often both done well, but a small number of candidates did not include the mass when forming their equations.

Q21.

There were many good responses to this question. Parts (a)(i) and (a)(ii) were often done well by candidates. A small number of candidates did not include the mass when forming their equations. In part (b), some candidates gave the velocity of the particle as the answer instead of the speed.

Q22.

Part (a) was done well by many candidates, although some assumed that the particles had coalesced. Some candidates produced equations with the correct terms, but the incorrect signs. For example, an equation such as $2m - 6 = 0.5m - 1.5$ was quite a common error.

Some candidates did part (b) very well and gained full marks. However, solutions where only one possible value for m was obtained correctly were seen quite frequently.

Q23.

Almost all the candidates were able to use the principle of conservation of linear momentum to find the velocity of A immediately after collision for part (a). However, some candidates lost accuracy marks due to mistakes made in collecting and simplifying the **i** and **j** terms. The most widely used method for part (b) was finding the two angles by using the inverse tangents and then adding the angles. Some candidates used a scalar product method.

For part (b), the great majority of the candidates understood that the impulse exerted by B on A was equal to the gain in the momentum of A. However, some candidates made sign errors by treating the gain in momentum of A as a loss of momentum. Having found the requested impulse, some candidates then proceeded to find its magnitude, which was not needed. Many candidates understood that the direction of the line of centres of the spheres was the same as the direction of the impulse.

Q24.

The candidates produced many good solutions to both parts of this question, with a high proportion of the candidates gaining full marks. There were three main errors that were observed on the scripts. Some candidates did not realise that one of the particles had a negative velocity before the collision and produced equations that would have applied if the particles had both been moving in the same direction. A few candidates found the momentum after the collision, but did not divide this by the appropriate mass to find the velocity. Some candidates made algebraic or arithmetic errors. For example in part (a), some candidates gave the answer 1.25, obtained by dividing 5 by 4, instead of the correct answer of 0.8.

Q25.

There were many good solutions to this question. The main issues that emerged were that some candidates found the momentum after the collision, but did not use the mass to find the velocity after the collision. There were also some candidates who did not know how to find the speed, who sometimes gave their answer to part (b) as a vector. The negative signs in the velocities caused some candidates to make arithmetic errors.

Q26.

The main pitfall for candidates in part (a) was a sign error in applying the principle of conservation of linear momentum. Candidates who made this error had not taken into account the opposite direction of motion of the two spheres before collision. Part (b) of this question was well-answered by the candidates.

Q27.

Candidates were able to construct appropriate momentum equations in both parts of the question, but errors with signs in part (b) were frequent.

Q28.

Despite being unusual this question was completed well by many candidates. Some candidates who struggled with algebraic expressions were subsequently very comfortable when able to substitute a numerical value for 'rn' Overall the handling of vectors was much better here than in Question 5. The most frequent errors included giving the final mass as '0.2m', including 'g' in momentum terms, and assuming the initial total momentum to equal zero. Confusion with scalar quantities was pleasingly rare.

Q29.

Part (a) This part was answered well by the great majority of candidates. However, a small number of candidates committed sign errors in using Newton's experimental law.

Part (b) Most candidates were able to apply the principle of conservation of the linear momentum and Newton's experimental law correctly but a significant number could not arrive at the correct quadratic equation in e. Most candidates solved the quadratic equation by using the formula and others by completing the square.

Q30.

Candidates produced many good solutions to this question. Most of the errors that were found were due to poor arithmetic or sign errors. A few candidates omitted the mass from one or more of the terms and occasionally a candidate would use weight instead of mass.

Q31.

This question was generally done very well and there were many complete correct solutions. The main source of errors was due to mixing up velocities and masses.

Q32.

Many candidates were able to make a reasonable attempt to produce a vector equation using conservation of momentum. A common error was to take the combined mass after the collision as $3m$ rather than $3 + m$. Given a vector equation some candidates were not able to find the mass, but there were also many good solutions. Surprisingly, the candidates seemed to find it easier to find V, than to find m.

Q33.

Part (a) of the question was answered correctly by many candidates. Some candidates made sign errors when they applied Newton's experimental law.

For part (b), most candidates applied the Impulse/Momentum principle efficiently to the sphere B rather than to the sphere A to show the result.

In part (c), a significant number of candidates went through long algebraic manipulations to find the result, and in doing so, they made mistakes.

Candidates could not gain any marks for part (d) without first substituting $e = \frac{4}{5}$ to find the velocity of A in terms of u.

Q34.

Part (a) produced many accurate solutions. Many were also successful in part (b) although some selected incorrect velocities for their calculations.

Q35.

Candidates are well versed in the use of conservation of momentum and Newton's law of restitution. Several candidates failed to take into account the sign of particle Q, thus obtaining $3mu$ for the initial momentum. Once the equations were set up they were generally solved correctly, although a few candidates forgot to find the speed of Q after the collision and careless algebraic slips sometimes cost a mark at the end. Part (a)(ii) was not well attempted with only the most able candidates being able to set up and solve the required inequality. In part (b)(i) the last mark was often dropped for not explaining why a negative sign had disappeared. This generally happened if the candidate considered the change in momentum of P rather than Q. The very last part seemed to puzzle a number of candidates who could clearly not see how the result followed by using $0 \le e < 0.5$.

Q36.

This proved to be a good opening question for most candidates with only occasional errors in algebraic manipulation.

Q37.

There was a very good response to this question. When errors were made they tended to be one of the following: not using different signs for the velocities and therefore getting 240 Ns; failing to realise that the force was a symmetrical curve; not labelling the axes or failing to show 1.2 on the t axis; or not realising that area = impulse in part (iv). In part (c) many candidates used calculus methods correctly. Equally, using the fact that the curve was symmetrical with t = 0.6 substituted to get 900N scored full marks.

Q38.

This question proved to be an excellent starter. Many candidates clearly knew the relevant formulae well. Almost all candidates scored full marks on part (a). Some candidates incorrectly tried to use $F = ma$ to solve part (b).

Q39.

Some candidates seemed to lose the thread of this question along the way. However, a better response than expected was made to part (d).

Part (a) was done very successfully, although sometimes an error in working meant that the required speeds were interchanged. In part (c) some candidates used the restitution equation again, but to no avail.

There were a variety of responses to part (d). These included: use of distance = speed × time to build up the solution in stages; use of the total time for both spheres being the same to form a single equation and attempt to solve it; and use of ratios related to distance = speed × time. The length of the solutions varied from two lines (fully correct) to more than two pages with a minor error along the way. The last part was meant to discriminate and it largely resulted in 0, 2, 5 or 6 marks: either a start was made and not seen through or the solution was almost fully correct.

Q40.

Part (a) was mostly done well although some inaccuracy crept in with candidates who reduced the given time to decimal form, resulting in failure to complete part (ii) correctly.

Part (b) brought a variety of responses, where some assumed the velocity of Q to be zero at the time of impact, while others used '9.8' for this value. Few realised the need to calculate this velocity, and there were other errors reflecting lack of appreciation of the directions of motion of the particles.

Q41.

There was an excellent response to this question. The great majority of candidates set up a pair of simultaneous equations and solved them. It was pleasing to see candidates showing sketches of the collision with velocities clearly labelled. Some candidates chose to use the answer given to find the other speed. This approach received either three or five marks depending on whether both equations had been set up or not. A few thought that Newton's law of restitution was $v + w = e(12 + 8)$,
or $(12 – 8) = e(v – w)$.

Q42.

There were many careless errors in part (a), especially with signs. Part (b) proved more discriminating than expected, and of the more valid attempts, again there were many sign errors.

Q43.

Candidates did not find this question easy. The principle of conservation of momentum was well understood, but there was little consideration of the directions of motion of the particles and sign errors were frequent in equations. Some poor algebra was seen, as was an occasional tendency to state and use a value for m.

Q44.

The required equations were usually set up well. A mark was sometimes lost in part (a) because of a numerical slip. Candidates who used the answer given to find the speed of A were penalised for not solving a set of simultaneous equations. Candidates who were not familiar with the correct units could not fully complete part (b). Candidates who found the impulse of B on A had to make clear reference to “equal and opposite impulses” to gain the final mark for part (b).

In part (c)(i) a few candidates thought that the graph had to be curved and not composed of two straight line segments. Part (c)(ii) proved to be a test of remembering a formula for the area of a triangle more than anything else. The quality of the criticism in part (c)(iii) showed a distinct improvement over previous papers, with candidates often indicating a sound understanding of the situation.

Q45.

This question showed considerable improvement in the manipulation of vector expressions. A variety of methods were used in part (a). Many candidates found the change in displacement and then divided by the time difference to find the velocity. Some attempts at this method showed only sparse working out, and candidates should be reminded that insufficient working out can lead to marks being lost. Other methods involved integration of the velocity to find and then use the equation of the path of the particle. It was gratifying to see such a sound understanding of mathematics on such scripts. Part (b) was done well, but part (c) proved more testing, with only a few realising the significance of the initial position of the particle.

Q46.

Although work with vectors has improved considerably in recent sessions, there is still some misunderstanding of the differences between vectors and scalars. Part (a)(i) was often successful, but common errors included the reversal of the vectors when subtracting and the use of incorrect units for momentum. Part (a)(ii), an unusual request, was quite well understood. Part (b) proved popular and successful, with many candidates using their result from part (a), while others reverted to using the conservation of momentum.

Q47.

This question proved to be more demanding than the first two questions. A number of candidates did not grasp the significance of the statement that the two particles were travelling towards each other and did not include a negative sign with one of the velocities.

In part (b) some calculated the speed of A rather than the speed of B.

Q48.

Part (a) was answered well.

In part (b) a few candidates only included the (unknown) number of arrows in one side of the conservation of momentum equation, while a few others found the final velocity by considering each of one, two three, four, five… arrows being fired, until they reached ten arrows which gave the correct result.

Q49.

This question was found to be a suitable start to the paper with the majority of candidates being successful.

Q50.

This question was done very well by many candidates. The main error was to ignore the fact that the particles were moving in opposite directions before the collision and to omit a negative sign from one of the velocities. A few candidates made minor arithmetic or algebraic errors when obtaining the final velocity.

Q51.

This question was answered well. A few candidates made errors with signs, as initial and final diagrams were rarely shown.

Q52.

This was also done very well and again many candidates gained full marks. A very small number of candidates did not know how to approach this problem and a few made arithmetic errors.

Q53.

Parts (a) and (b) were completed well with only a few arithmetical slips being made, the most common of which was dropping a minus sign. A significant proportion of candidates did not appreciate that the direction of the line of centres was the same as the direction in which the momentum was changed.

Q54.

There were many very good responses to parts (a) and (b). There were a few candidates who used the product of weight and velocity rather than mass and velocity. There were far fewer good responses to part (c). Many candidates started this part by applying conservation of momentum, rather than forming an inequality based on the motion of the particle.

Q55.

Part (a) was answered badly; many failed simply to multiply $2u$ by $3m$, the mass of Q. Part (b) was usually answered correctly. Candidates found the proof of $M \geq m$ challenging; many assumed it to be clear from $mu = Mv$. Justification was required, using $e \leq 1$, or using the fact that kinetic energy does not increase in a collision.

Q56.

This question was generally done very well. A few candidates made errors when calculating the total mass of the moving trolley, child and ball. A very small number of candidates used weight instead of mass.

Q57.

The majority of candidates also did this question very well. Almost all candidates completed part (a) correctly. Part (b) was done well, the only common error being to omit the negative sign needed for the initial velocity of C. Part (c) was also done well. A few candidates did not realise how to approach this part, some making no attempt and others producing confused working. In this question a few candidates used weight instead of mass.

Q58.

A large number of candidates correctly used conservation of momentum. Those who did not use a clear 'before and after diagram' showing the directions of velocities, often used creative sign changes to obtain a positive value for e.

Q59.

Most candidates readily used the fact that the velocity parallel to the wall was unaltered in the impact. They also considered velocities perpendicular to the wall although with less success. Occasionally the signs used in this restitution equation were incorrect. Thus, many candidates correctly found the angle of 52.4° being the angle with which the particle left the wall. However a number did not then find the required angle to be 112.4°. In part (b), most candidates did not use the fact that the impulse (acting perpendicular to the wall) is the change in momentum **perpendicular** to the wall. Most candidates correctly multiplied the change in momentum by 20 (being 1/time) to find the impulsive force.

Q60.

Surprisingly part (a) was often incorrect; $3mu$ was a common answer. Again, in part (b) for those candidates who did not use a diagram, signs were often incorrect, particularly in the equation using coefficient of restitution. Those who did use the correct signs had little difficulty in parts (b) and (c). The majority of those who appreciated what was happening in part (d) were also successful. Many found the position of A when B hit the wall. Using the ratio of the final speeds was found to be a simple method of finding the distance of the collision from the wall.

Q61.

This question caused major problems. Conservation of momentum over the whole system for both collisions produced, for most candidates, the common velocity required in part (a). The problem which arose in part (b) was that candidates confused the impulse in each of the two strings with the resulting impulse on each of the three spheres. The particle B subject to two impulses caused concern to many candidates. The more able candidates just used the impulses on particles A and C, which readily gave the impulses in each string.

Q62.

The candidates found this question straightforward and produced many good solutions. A few candidates did not use the correct total mass in part (b), omitting the mass of one of the pellets.